

APPLE

UI QUALITY ENGINEERING · FROM USER EXPERIENCE TO CONTINUOUS QUALITY IMPROVEMENT

DAY 40/100 · 05 SEPTEMBER 2026 | Jagriti Singh | Quality Engineer

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01 — CASE OBJECTIVE

This case study explores how Quality Engineering principles can be embedded into UI/UX design to create interfaces that are not only visually clear, but also reliable, user-centred, measurable and continuously improvable.

The proposed framework treats the user interface as a quality system. Instead of considering UI quality only in terms of visual appearance, the approach focuses on helping users make informed decisions, preventing avoidable errors, enabling recovery when failures occur, validating design changes through evidence, and continuously improving the experience through user feedback and behavioural signals.

02 — QUALITY PROBLEM

How can UI/UX be designed and managed as a measurable quality system where user decisions, error prevention, recovery, usability and continuous improvement are treated as interconnected quality requirements?

USER DECISION SUPPORT

ERROR PREVENTION

ERROR RECOVERY

USABILITY VALIDATION

CONTINUOUS QUALITY

03 — USER DECISION SUPPORT

Instead of relying only on a simple “Are you sure?” confirmation message, the interface provides relevant information or contextual questions before a critical action.

INFORMATION → UNDERSTANDING
→ USER DECISION → ACTION

CRITICAL ACTION CONFIRMATION

Permanent storage changes will affect
all synced devices immediately.

Confirm & Proceed

The user remains in control rather than being
pushed toward a predetermined choice.

04 — ERROR PREVENTION & RECOVERY

If an unexpected error occurs during an important action, the proposed interface first attempts automatic recovery wherever technically possible. The user receives clear feedback about the current state.

AUTOMATIC RECOVERY → CLEAR FEEDBACK → USER-
CONTROLLED RECOVERY

RECOVERY STATE LOGIC

Retry

Cancel

Recover Session

05 — UI QUALITY AS AN ENGINEERING PROCESS

USER REQ

UI/UX
DESIGN

PROTOTYPE

USABILITY

FRICTION
ID

ROOT
CAUSE

IMPROVEMENT

VALIDATION

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UI QUALITY ENGINEERING · FROM DESIGN TO QUALITY

06 — VALIDATING UI QUALITY

A design should not be considered successful simply because it looks cleaner or more modern. The proposed framework validates improvement through measurable outcomes.

-34%

TASK TIME

99.2%

SUCCESS RATE

07 — A/B TESTING

Comparing alternative interface designs using defined quality criteria across completion time, error frequency, user behaviour and usability outcomes.

Design A: Baseline

Design B: Proposed

08 — USER SEGMENTATION & KANO

User feedback is segmented to identify common patterns and underlying requirements. Kano perspective classifies must-be expectations from performance attributes.

Must-Be

Performance

Attractive

09 — CONTINUOUS QUALITY LOOP

Validated improvements are incorporated into design standards and future development practices through a continuous feedback and monitoring loop.

STANDARDISE → MONITOR → SIGNALS → DEVIATION → IMPROVE → RE-TEST

10 — USER BEHAVIOUR SIGNALS

Real user behaviour provides additional signals about interface quality. Continuous monitoring helps identify repeated interaction difficulties, recurring errors, and high-friction journeys before they become major usability issues.

11 — QUALITY ENGINEERING × UI/UX

QE: Measurement · Root Cause · Validation · Risk Thinking

UI/UX: Interaction · Usability · Accessibility · Experience

Result: User Need → Design → Measure → Analyse → Improve

KEY QUALITY INSIGHT

"UI quality is not only about how an interface looks. It is about how reliably it helps users make decisions, complete tasks, prevent errors, recover from failures and continuously improve through evidence."

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